



KCL AND KVL

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona
Electronics Engineer

TOPIC OUTLINE

Circuit Convention

Kirchhoff's Current Law (KCL)

Kirchhoff's Voltage Law (KVL)



CIRCUIT CONVENTION



CONVENTION

A convention is a widely accepted practice, method, or behavior that is followed by common **agreement** or tradition, rather than by formal rules.

Color coding in Offices

red – urgent documents

blue – general files

green – financial records

This is a common practice but not formally regulated.



STANDARD

A standard is a formal, established guideline, rule, or specification that is often **mandatory** and enforced by an authoritative body or organization.

IEC 60062 Resistor Color Code

black – 0

brown – 1

red – 2

.

.

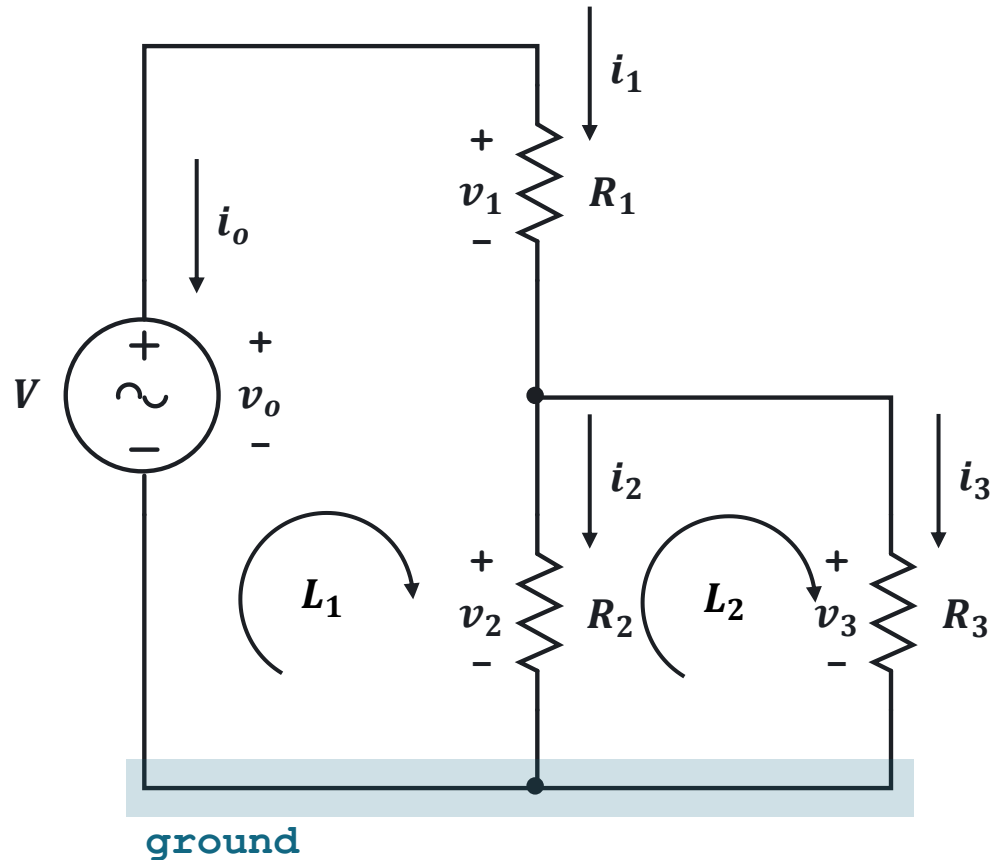
.

white – 9

Resistors have colored bands that represent specific digits, multipliers, and tolerance values.



LABELING VARIABLES



Steps in Labeling Variables

1. Label the Reference Node (ground):

Select a reference node with the most connections or the negative (-) terminal of a voltage source.

2. Label Node Voltages:

Mark higher potentials as positive (+) relative to the reference node.

3. Label Currents:

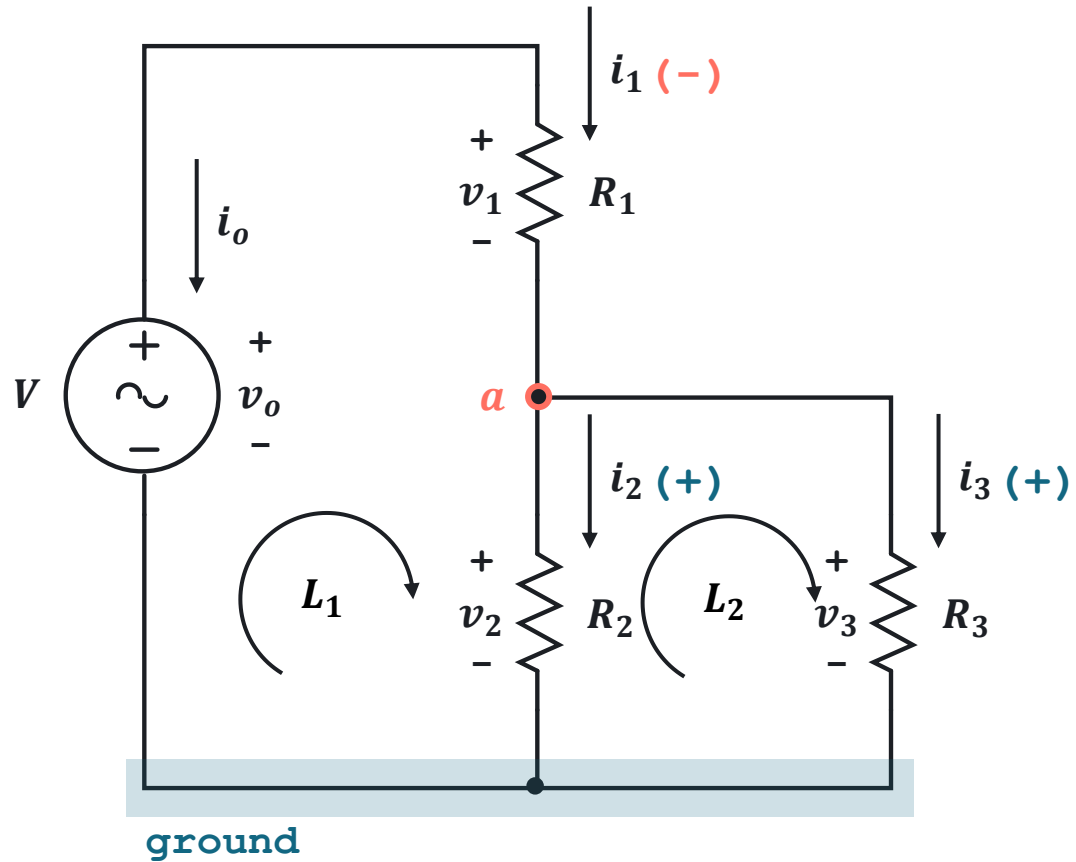
Entering the positive (+) terminal of a component.

4. Create a voltage loop:

Follow the defined current directions.



CURRENT FLOW CONVENTION



Current Flow Convention

- Current **entering** a node is negative **(-)**
- Current **leaving** a node is positive **(+)**

@ a

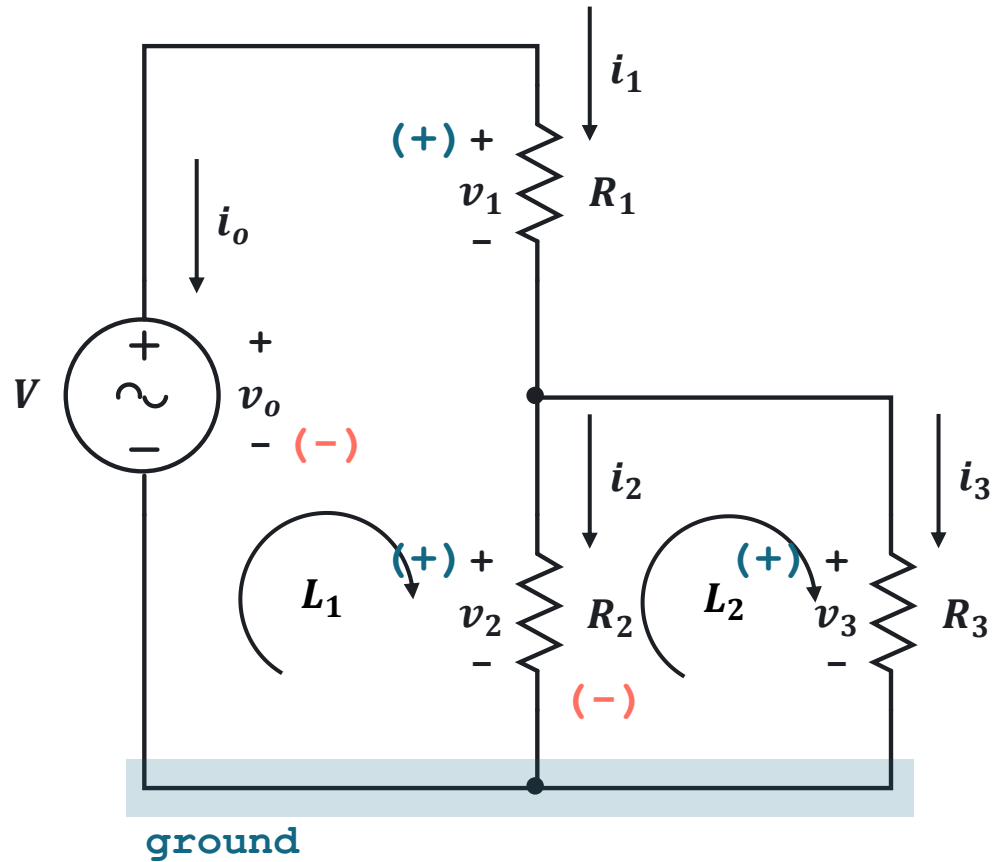
$-i_1$

$+i_2$

$+i_3$



VOLTAGE LOOP CONVENTION



Voltage Loop Convention

The “**sign**” of voltage of the element is the **first sign** the loop encounters.

@ L_1

$-v_0$

$+v_1$

$+v_2$

@ L_2

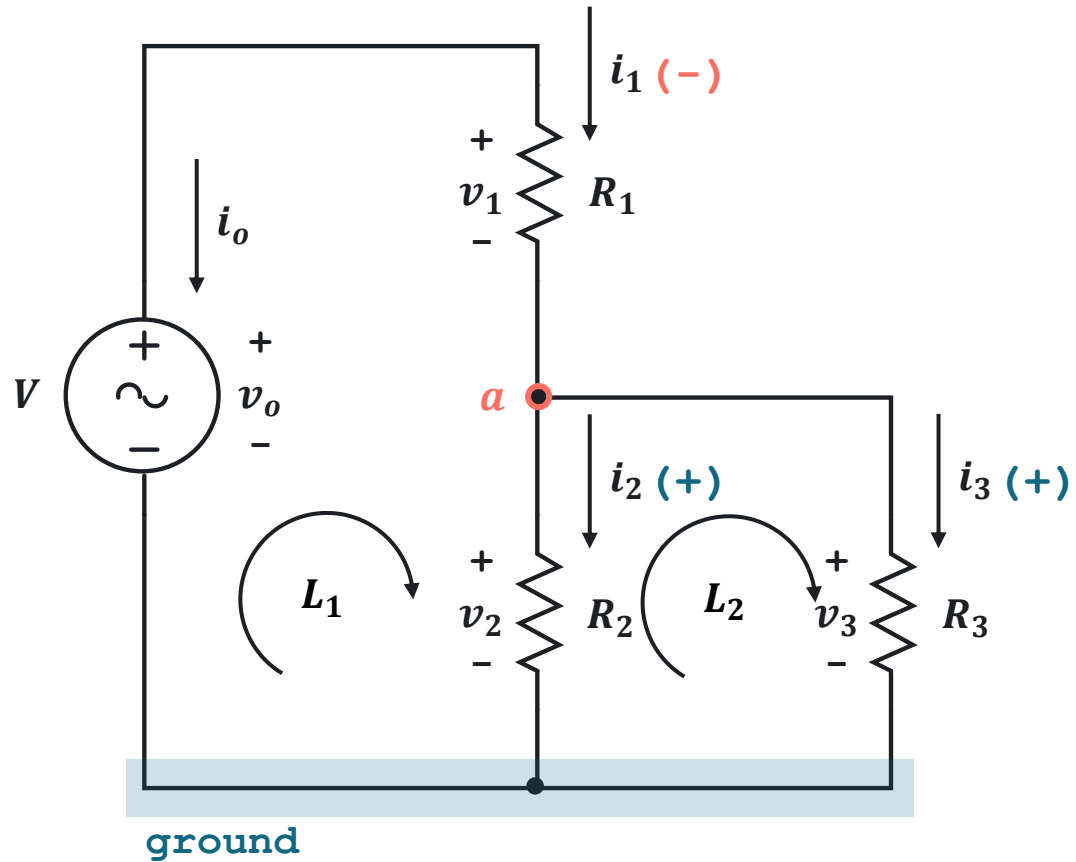
$-v_2$

$+v_3$



KIRCHHOFF'S CURRENT LAW AND VOLTAGE LAW





Kirchhoff's Current Law

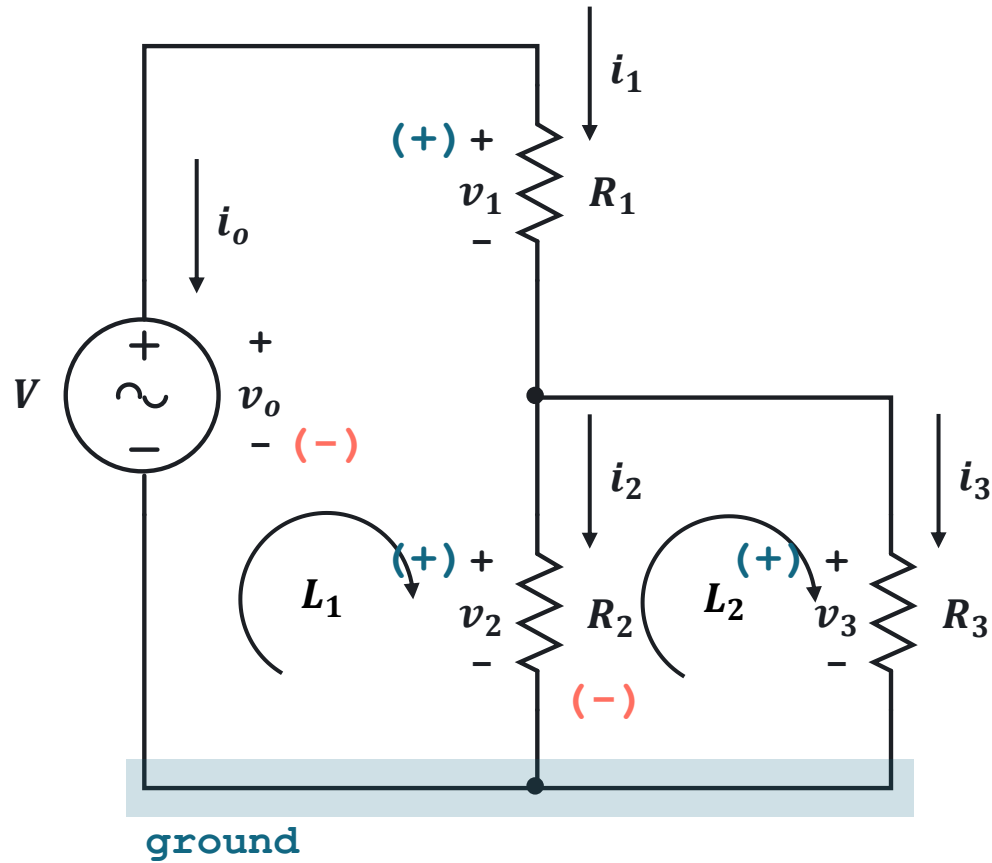
The summation of currents going-in and going-out a node is zero.

$$\sum i_j = 0$$

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$





Kirchhoff's Voltage Law

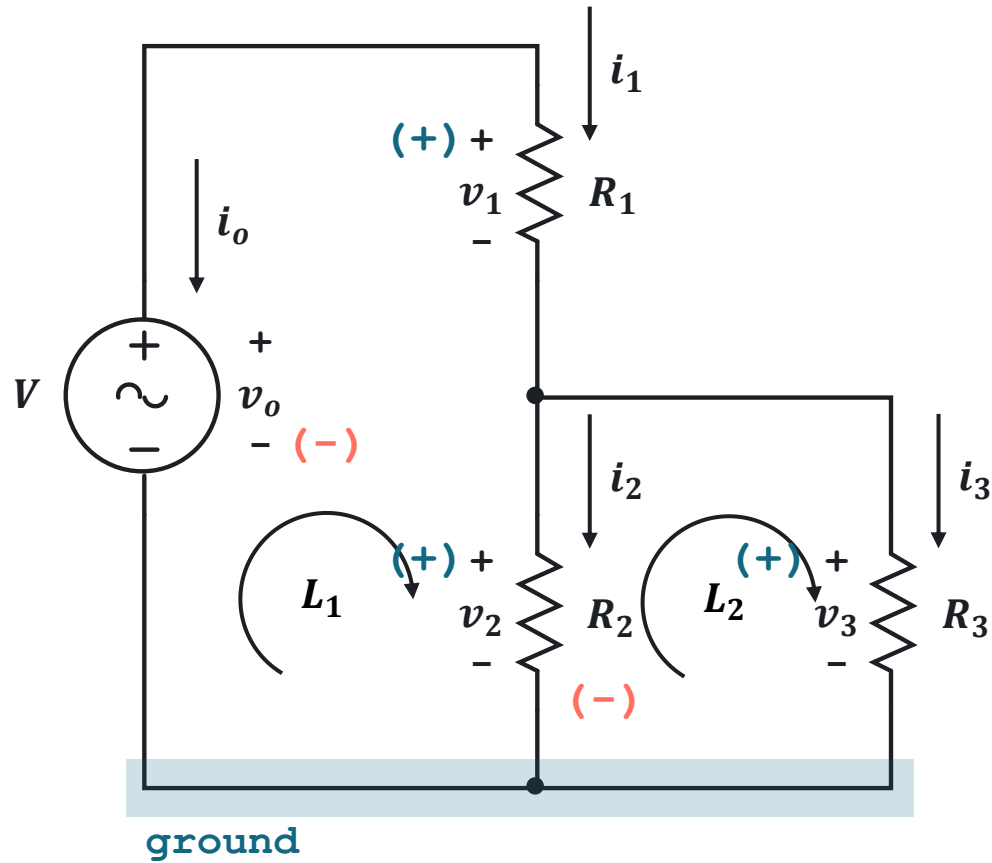
The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @ L_1

$$-v_0 + v_1 + v_2 = 0$$





Kirchhoff's Voltage Law

The summation of voltages in a closed-loop is zero.

$$\sum v_j = 0$$

KVL @ L_2

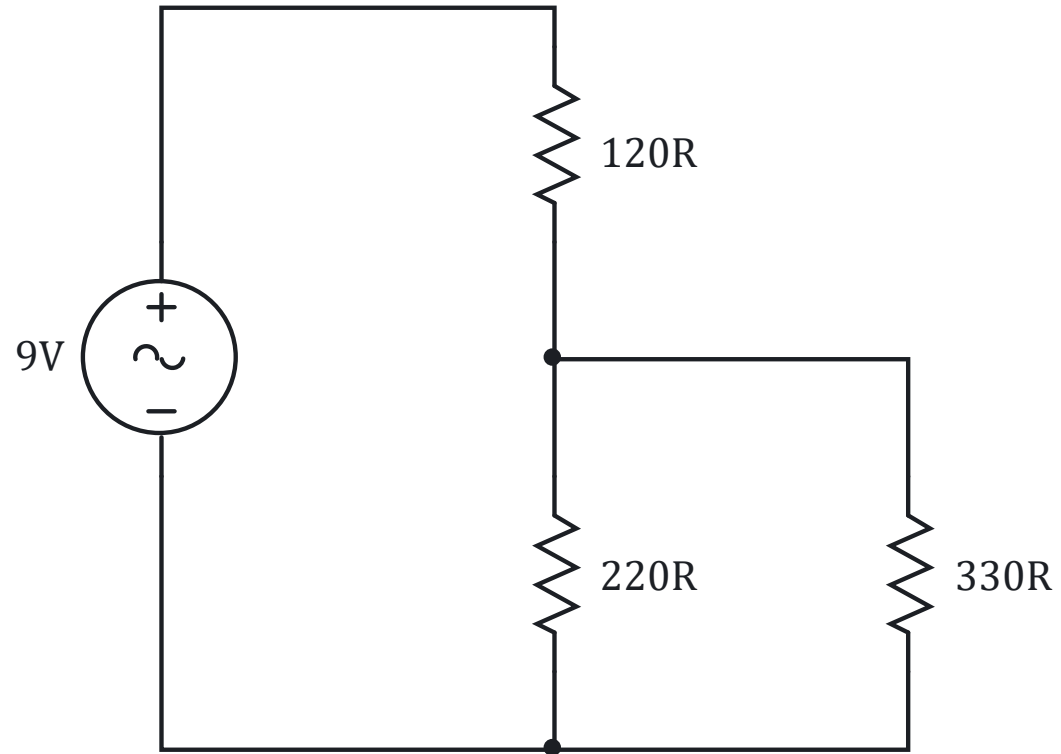
$$-v_2 + v_3 = 0$$



EXERCISE

Determine the current flowing through each resistor and the voltage drop across each resistor in the given circuit.

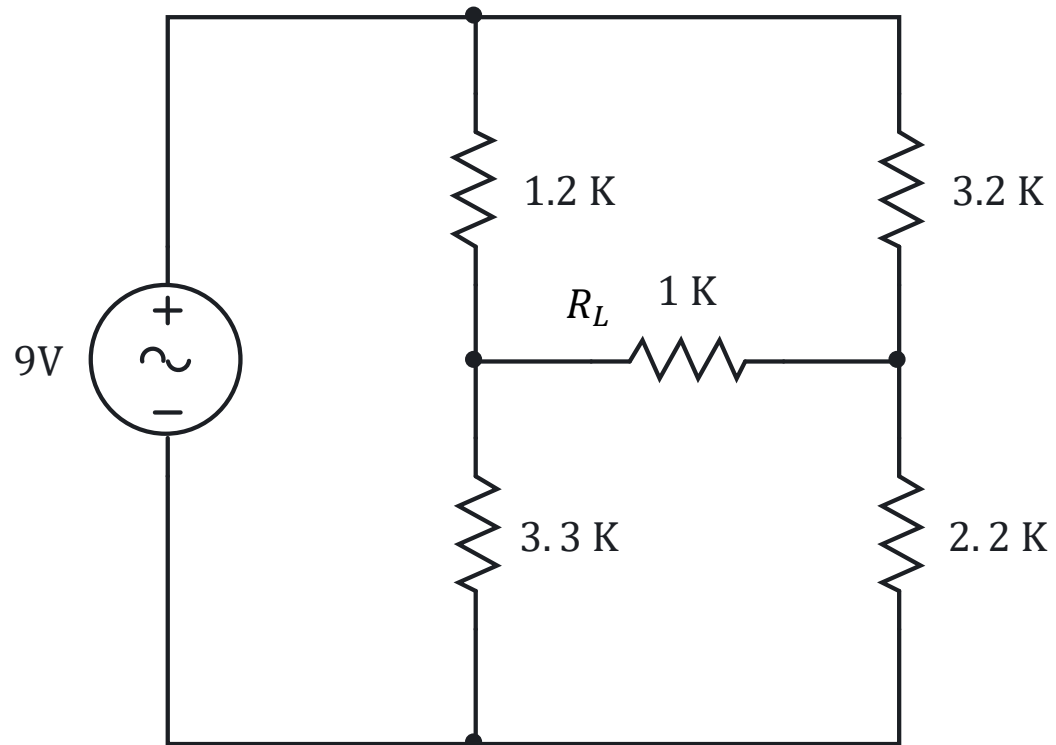
Solution



EXERCISE

Determine the load R_L voltage and current of the given circuit.

Solution



LABORATORY

